Archived Web Pages & the Wayback Machine

Week 7 · Documents module · Textbook Chapter 7

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This week at a glance

The web is not permanent. This week we scrape the *past* — and measure how pages change over time.

Monday | Concepts & notebook intro

- Link rot, the Internet Archive, and two APIs into the past

Wednesday | Notebook lab

- Work the Chapter 7 notebook: Availability API, CDX, tracking a Terms-of-Service document

Friday | Textbook revisions

- Propose & peer-review pull requests improving Chapter 7

Companion notebook

`ch-07-archives.ipynb`

Bring a laptop

Wednesday is hands-on — we query the Archive live.

1. Monday · Concepts

The web is not permanent

Pages change, disappear, and get redesigned constantly. A large fraction of URLs break within a few years — **link rot.**

This is a research problem: how do you study something that might not be there tomorrow, or measure how it *changed* when only today's version is visible?

The **Internet Archive** (Brewster Kahle, 1996) answers by archiving the web. Its **Wayback Machine** has captured over 800 billion pages, free to anyone.

It is both a data source and *counter-erosion infrastructure* — a community-governed public resource (the book's "post-API" ownership theme).

The Wayback calendar: every dot is a capture you can retrieve.

Textbook Ch. 7, "The Web Is Not Permanent."

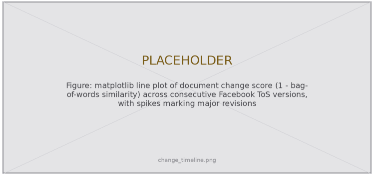
From snapshots to a longitudinal source

One old snapshot is a curiosity. **Comparing snapshots across time** is where the research value lives.

The Wayback Machine turns the web from a snapshot medium into a **longitudinal data source** — you can *measure* change, not merely observe it:

- Track a platform's Terms of Service across a decade
- Follow a government page across administrations
- Contrast two rivals' homepages as they evolve

You are not just reading old pages — you are measuring **institutional behavior over time.**



A "change timeline" we build Wednesday: spikes mark major revisions.

Textbook Ch. 7, "Comparing Multiple Snapshots" & "Tracking Policy Changes."

Raw or wrapped? The `id_` rule

A snapshot URL looks like:

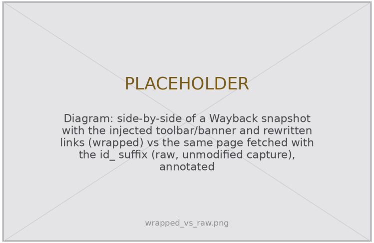
`web.archive.org/web/20000815052826/http://cnn.com/`

By default the Archive **injects** its own toolbar, banner, and JavaScript, and **rewrites** every link, script, and image to point back into the archive. Convenient for browsing — **poison for measurement:** your link, script, and image counts include the scaffolding.

Append **`id_`** to the timestamp

`.../20000815052826id_/http://cnn.com/`

to get the **original, unmodified capture.** Raw doesn't render prettily — but for measurement, raw is exactly what you want.



Wrapped (toolbar + rewritten links) vs. the **`id_`** raw capture.

Textbook Ch. 7, "The Availability API." See also Brügger (2019) on archived-web caveats.

What the notebook covers

This week's companion notebook builds one skill at a time:

- **Availability API** — find the snapshot closest to a date; retrieve it raw with `id_`
- **CDX Server API** — enumerate a URL's *entire* capture history, filtered server-side
- **Policy tracking** — loop over quarterly dates, collect a document's history, serialize to JSON
- **Bag-of-words** — tokenize, drop stopwords, and measure how much consecutive versions changed

Connecting to Ch. 6 & what's next

Archived pages parse with `BeautifulSoup` exactly like live ones (Ch. 6), and the same `time.sleep()` politeness applies — the Archive is a nonprofit. The JavaScript growth we measure is *why* dynamic-page techniques (next week) exist.

2. Wednesday · Notebook Lab

Finding a snapshot — the Availability API

The simplest question: does a snapshot exist near a date? Pass query args as a `params` dict so `requests` URL-encodes them (one value is itself a URL):

```
import requests

api_url = "https://archive.org/wayback/available"
response = requests.get(api_url, params={"url": "https://www.cnn.com",
                                         "timestamp": "20000815"}) # Aug 15, 2000
data = response.json()

if data["archived_snapshots"]:
    snap = data["archived_snapshots"]["closest"] # closest capture to that date
    print(snap["timestamp"], snap["url"])
else:
    print("No snapshot available near this date")
```

Textbook Ch. 7, “The Availability API.”

Retrieve it raw with `id_`, then measure

Rewrite the snapshot URL to the `id_` form, then parse it like any page:

```
from bs4 import BeautifulSoup

ts = snap["timestamp"]
raw_url = snap["url"].replace(ts, ts + "id_") # id_ -> original, unwrapped capture

soup = BeautifulSoup(requests.get(raw_url).text, "html.parser")
stats = {"year": ts[:4], "num_links": len(soup.find_all("a")),
        "num_scripts": len(soup.find_all("script"))}
```

The same measurement across three eras tells the web's story:

```
# year num_links text_length num_images num_scripts
# 2000 150 8500 20 2 # largely static HTML
# 2024 800 45000 120 95 # a JS app that renders
```

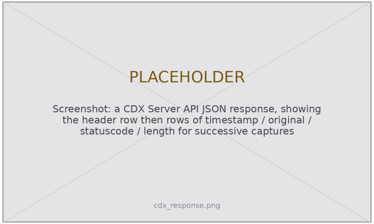
Textbook Ch. 7, "Comparing Multiple Snapshots." Without `id_`, every count is inflated by the toolbar.

The whole history — the CDX Server API

Availability finds *one* snapshot; CDX enumerates *all* of them. Two params filter server-side — less data, less cleanup:

```
cdx = "https://web.archive.org/cdx/search/cdx"
params = {"url": "facebook.com/terms",
          "output": "json",
          "fl": "timestamp,original,statuscode,length",
          "filter": "statuscode:200", # only 200s
          "collapse": "digest"} # drop repeats
data = requests.get(cdx, params=params).json()

df = pd.DataFrame(data[1:], columns=data[0])
df["timestamp"] = pd.to_datetime(
    df["timestamp"], format="%Y%m%d%H%M%S")
```



Row 0 is the header; each later row is one distinct capture.

Textbook Ch. 7, "The CDX Server API." Add `"limit": 100` while developing.

Tracking a document over time

Wrap the Availability + `id_` + parse steps in `get_wayback_content()` (returns `None` on a miss), then loop over quarterly dates:

```
dates = pd.date_range(start="2005-01-01", end="2024-01-01", freq="QS")

results = []
for date in dates:
    snap = get_wayback_content("facebook.com/terms", date)
    if snap is not None: # skip misses instead of crashing the run
        results.append(snap)
    time.sleep(1) # rest between requests -- the Archive is a nonprofit
# Retrieved 68 snapshots from 77 time points
```

Serialize immediately — you may not be able to re-scrape it later:

```
import json
with open("fb_tos_history.json", "w") as f:
    json.dump(results, f, indent=2) # (also cache each raw capture as you go)
```

Textbook Ch. 7, "Retrieving Historical Content," "Serializing," & "Caching Raw Captures."

Measuring change — bag-of-words

Represent each version as a vector of word frequencies (order ignored), then compare vectors:

```python
from gensim.utils import simple_preprocess
from gensim import corpora, similarities
from nltk.corpus import stopwords # nltk.download("stopwords") once

stop = set(stopwords.words("english"))
docs = [[t for t in simple_preprocess(r["content"]) if t not in stop] for r in results]

dct = corpora.Dictionary(docs) # word -> id
corpus = [dct.doc2bow(d) for d in docs] # each version -> word-count vector
index = similarities.SparseMatrixSimilarity(corpus, num_features=len(dct))
change = [1 - index[corpus[i-1]][i] for i in range(1, len(corpus))] # 0=same, 1=all-new
```

Plotting `change` over time is the timeline from Monday: spikes flag major revisions — a new policy, a redesign, a response to controversy.

Textbook Ch. 7, “Document Similarity with Bag-of-Words” & “Visualizing Document Change.”

Lab: work these, then show-and-tell

Work through the notebook exercises in pairs:

1. Track a policy document (privacy, academic-integrity, agency FAQ) across ≥ 5 time points; plot length
2. CDX: compare capture frequency & page size for two rival organizations
3. Pairwise similarity across *all* versions $\rightarrow$ a heatmap of turning points
4. Archival history of a government site — gaps? coverage? size drift?
5. Then-and-now: earliest vs. latest snapshot, 200-word diff analysis
6. **5617**: assess Wayback's limits (toolbar, capture bias, gaps) vs. Rogers (2019)

Show-and-Tell

Come ready to share:

- a challenging bug
- a change you found in the archived past
- a provocation for a research design

You are not reading old web pages.
You are measuring how institutions behaved —
one archived capture at a time.

The page you study today may be gone tomorrow:
serialize everything.

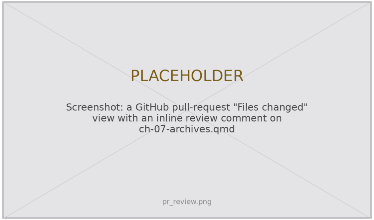
3. Friday · Textbook Revisions

Improving Chapter 7 together

Today we make the book better. Each of you opens a **pull request** against `ch-07-archives.qmd` and reviews a classmate's.

The workflow (from Week 1):

1. Branch / fork the book repo
2. Edit the chapter; commit with a clear message
3. Open a PR describing *what* and *why*
4. Review two peers' PRs: kind, specific, actionable



Contributions & reviews count toward the Textbook Revisions grade (15%).

A revision menu for Chapter 7

High-value targets — pick one and scope it to a single PR:

- **Run it against the live APIs.** The comparison numbers and "68 of 77 snapshots" are *illustrative*. Query the real Availability/CDX endpoints, report actual results, and label sample outputs as representative.
- **Add the chapter's first figures.** It is all code today: a Wayback calendar view, an annotated wrapped-vs-`id_` capture, or a CDX-response schema would anchor the APIs.
- **Close a gap in "Common Issues to Debug."** A worked `id_` example (link counts wrapped vs. raw), a copy-paste `nltk.download("stopwords")` recipe, or a CDX missing-field case.
- **Make an exercise self-checking.** Add expected output, a rubric, or a starter cell to the open-ended policy-tracking and heatmap exercises.
- **Promote the limitations into the main text.** Capture-frequency bias, coverage gaps, and link rewriting currently live only in the graduate exercise — tie them to "Further Reading" (Brugger et al. (2018)).

Targets drawn from the chapter's callouts, "Common Issues," exercises, & "Further Reading."

What makes a good revision & a good review

A strong PR

- One focused change, clearly titled
- Explains the problem it fixes
- Builds (`quarto render`) without errors
- Matches the book's voice and notation
- Discloses any AI assistance

A strong review

- Runs / reads the change, not just skims
- Names specifics (line, claim, code)
- Separates "must fix" from "nice to have"
- Is generous — assume good faith
- Ends with a clear verdict

Next week: **Dynamic pages** — when the data appears only after JavaScript runs (Selenium).

Key takeaways

1. The web is impermanent (link rot); the **Internet Archive** is public infrastructure that preserves it.
2. **Availability API** finds one snapshot near a date; the **CDX Server** enumerates the full history — filter and collapse server-side.
3. Append `id_` for the **raw** capture; the toolbar-wrapped version pollutes every measurement.
4. Wayback + text analysis turns the archive into **longitudinal data** — measure change, don't just observe it.
5. Be a good citizen: `sleep` between requests, cache raw captures, and **serialize** — you may not be able to re-scrape.

Reading: Textbook Ch. 7. Related: Brugger et al. (2018); Rogers (2019).